SECJ 4383 – Software Constructions Lab Exercise

Lab Exercise 1a – DevOps Continous Integration Lab

(This lab exercise is not self-explanatory. Further explanation will be given by the instructor during the lab session)

Task 1: Install Spring Framework:

To install Spring Framework, you can follow the official documentation provided by Spring at https://spring.io/projects/spring-framework. This documentation provides step-by-step instructions on how to install and configure Spring Framework on your local machine.

Task 2: Create a Spring Framework project:

To create a new Spring Framework project, you can use Spring Initializr, which is a web-based tool that generates a starter project with the required dependencies and configurations. You can access Spring Initializr at https://start.spring.io/. After selecting the required dependencies and configurations, you can download the project and import it into your IDE.

Task 3: Install Git:

To install Git, you can follow the instructions provided in the official Git documentation at https://git-scm.com/book/en/v2/Getting-Started-Installing-Git.

Task 4: Create a Git repository:

To create a new Git repository, you can use a cloud-based Git hosting service such as GitHub or GitLab. You can create a new repository by following the instructions provided by the respective service.

Task 5: Integrate Git with your Spring Framework project:

To integrate Git with your Spring Framework project, you can use the Git plugin provided by your IDE. Most IDEs, such as Eclipse and IntelliJ, have built-in Git support. Alternatively, you can use the command-line interface to interact with Git.

Task 6: Install Jenkins:

To install Jenkins, you can follow the instructions provided in the official Jenkins documentation at https://www.jenkins.io/doc/book/installing/.

Task 7: Install Jira:

To install Jira, you can follow the instructions provided in the official Jira documentation at https://confluence.atlassian.com/adminjiraserver/installing-jira-applications-on-linux-938846841.html.

Task 8: Add JDK and Maven home path directory in Jenkins

In order to add JDK and Maven home path, you need to install maven and jdk plugin in jenkins. You can follow guide from official Jenkins documentation :
https://www.jenkins.io/doc/book/managing/plugins/

Task 9: Create a Jenkins job:

To create a new Jenkins job, you can follow the instructions provided in the official Jenkins documentation at https://www.jenkins.io/doc/book/pipeline/getting-started/. This documentation provides step-by-step instructions on how to create a new Jenkins job and configure it to build and test your Spring Framework project.

Task 10: Integrate Jenkins with GitHub

To integrate Jenkins with github, you can follow tutorial from this video: https://www.youtube.com/watch?v=-KrlCWVPfJM. This video shows step by step on how to integrate Jenkins with github

Task 11: Integrate Jira with your Spring Framework project in Jenkins:

To integrate Jira with your Spring Framework project, you can use a plugin such as Jira Integration for Jenkins. This plugin allows Jenkins to update Jira issues with build and test results. https://plugins.jenkins.io/jira/

Task 12: Test connection between Github and Jenkins

To test the connection, you can add github credential in Jenkins job configuration then click on the 'Build' button to check commits and console

Task 13: Test Integration between Jenkins and Jira

The integration can be done by making sure that both Jenkins and Jira already installed required plugins. You need to add 'post-build action' section by adding 'Jira:Create new issue'. Project information must be correct in order for the integration to occur. Click on the 'Build' button to check build status (if build failed, Jira issues will be created automatically).